

The Hydraulic Aspects of High Head Barrages

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Abstract:

Many barrages are built across the globe which are primarily used for irrigation, flood control and drinking water. The barrages are generally having low depth of flow over the crest, so as to keep the discharge intensities of the order of 50 m³/s/m or less for maximum probable flood. As the depth of flow increases, the velocities and discharge intensities considerably increase. In large rivers such as, Godavari, Ganga, Narmada, Yamuna and Kosi, multiple number of spans of the barrages are necessary to pass the flood discharges. In such cases it becomes necessary to provide depth of overflow over the crest of barrage as high as 10 to 12 m to accommodate the high flood discharges. For such depth of overflow, it is very complex task to design energy dissipation arrangement in view of the, velocities of the order of 15 to 18 m/s and discharge intensities more than 60 to 80 m³/s/m. It may be noted here that, many barrages already constructed in India and Abroad are facing many hydraulic issues for the efficient operation of the barrage due to high heads and high discharge intensities. Also, the schedule of operation for large barrages having more than 10 spans has to be evolved very judiciously so as to avoid unfavorable flow conditions such as circulation and return flows across the stilling basin as well as in the downstream river portion. The paper discusses the hydraulic design aspects regarding energy dissipation arrangement for such high head barrages. The prototype data of various barrages was reviewed in the light of above mentioned aspects. The alternative arrangements for achieving the efficient energy dissipater have also been elucidated in this paper. The results of the hydraulic model studies conducted at Infraplan Hydraulic Laboratory, Pune for a barrage having depth of overflow of 12 m over the crest and having 66 spans have also been presented.

Keywords: High Head Barrages, Discharge Intensities, Energy Dissipation.

1. Introduction

Barrages are gated structures built across rivers to regulate water flow and maintain water levels. Barrages play a vital role in water resource management, particularly for irrigation, flood control, and drinking water supply. Many of the barrages are seated over permeable foundation. Unlike dams, barrages allow for controlled releases while keeping the upstream water level stable, facilitating irrigation, navigation, and drinking water supply. As climate variability intensifies and water demand increases, the design and operation of barrages have become more complex. This paper addresses some of the hydraulic challenges associated with high-head barrages. Barrages generally feature low flow depths over their crests and design consideration is to limit the discharge intensities to around 50 m³/s/m during maximum probable floods. The design must ensure stability and prevent erosion while allowing for adequate passage of floodwaters. However, in practice for many barrages the discharge intensities are much higher and head over the crest are of the order of 10 m are in vogue. It's essential to address energy dissipation properly, as velocities leaving the crest could reach around 15 m/s. This necessitates addressing various hydraulic considerations with great intensity. The present paper highlights these issues and probable solutions for the layout of the barrages including the crest, energy dissipation, downstream protection works to the barrage and protection of the downstream alluvial river, which are not included in most of the guidelines and design procedures.

2. Design of Barrage Crest

Barrages are typically designed with an overflow depth of 4 to 5 meters and a discharge intensity of 40 to 50 m³/s/m. For such conditions, a flat crest is usually adopted to simplify construction and accommodate vertical lift gates. The width of the barrage crest ranges from 5 to 10 meters, depending on upstream foundation levels. However, as the head increases, the lower nappe of the water jet leaving the crest tends to separate from the downstream slope and strike the downstream apron, potentially causing damage as shown in Figure 1 and atypical example observed on the model is shown in Photo 1. For barrages with horizontal crests and overflow depths exceeding 10 meters, the resulting high velocities up to 15 m/s can lead to erosion of the concrete at impact points and compromise the effectiveness of energy dissipation systems as mentioned in the following paragraphs.

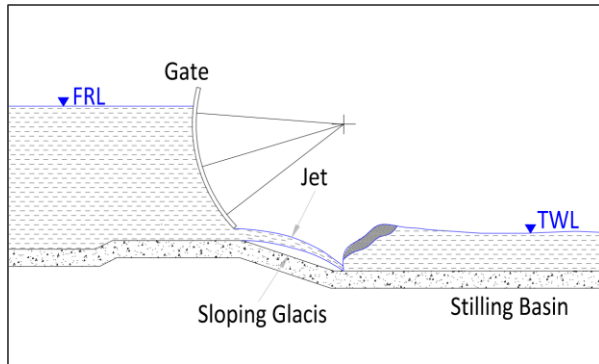


Figure 1: Jet hitting the downstream apron



Photo 1: Jet hitting the downstream apron –from model studies

To address these issues, it is essential to carefully design the barrage crest, particularly the downstream profile. The design should ensure that the water jet remains in contact with the downstream slope to prevent separation and improve stability. This is especially important for depth of overflows greater than 5 to 6 meters, as a well-designed profile helps in forming effective hydraulic jumps in the stilling basin. Proper crest geometry enhances pressure distribution and prevents pulsation of the jet due to fluctuations of the pressures beneath the water jet, improving the overall performance of the structure. This updated design approach is essential for reducing erosion risks and ensuring the barrage operates effectively under higher head conditions.

3. Energy Dissipation

Energy dissipation downstream of a barrage, as per BIS 4997:1968, is typically designed using a stilling basin, particularly for alluvial rivers. When flow velocities reach around 15 m/s, the hydraulic jump formation cannot be achieved for the entire range of gate openings. Additionally, deep scour holes created by high-velocity jets are unacceptable, as they pose a risk to the safety of the cut-off wall or sheet piles at the downstream end of the stilling basin. In the large width barrages, building up the tailwater level in the stilling basin takes time. During this period, the high-velocity jet can exit the stilling basin (with or without an end sill), potentially damaging the downstream protection works. To mitigate this, careful studies are needed to determine the appropriate depth and length of the stilling basin, based on the tailwater rating curve. Furthermore, the apron design must account for fluctuations in tailwater levels, which can vary by $\pm 10\%$ of the average river water level.

When the head increases, leading to higher incoming velocities in the stilling basin, conventional elements such as chute blocks and baffle blocks may not be effective. High fluctuating loads of 10 to 20 tonnes/m² can damage these blocks depending on the apron level. Numerous studies have been conducted on various sizes and shapes of chutes and baffle blocks, providing recommendations for effective energy dissipation, however, for high heads, experience indicated that, the proper functioning of the stilling basin is not achieved and shooting flows are observed downstream of stilling basin.

To address these issues, comprehensive hydraulic model studies are very much essential. These studies should determine the optimal gate operation schedule and the need for divide walls between groups of spans to prevent circulating flow currents within the stilling basin. The studies must also observe flow conditions for various combinations of reservoir levels and gate openings. It is crucial to prioritize hydraulic efficiency over cost optimization while designing the stilling basin to ensure proper energy dissipation and long-term structural safety.

4. Protection downstream of stilling basin

Generally, concrete block protection is provided downstream of the end sill of the stilling basin. In order to resist uplift forces if any, concrete blocks of various sizes are being laid arbitrarily. It may be mentioned here that, few blocks would resist maximum uplift pressure even if it gets launched. It is recommended to incorporate the consideration of stone velocity and weight into the design of downstream protection, ensuring that the size and weight of the concrete blocks or stones are based on the expected flow velocities, using empirical data and hydraulic studies to prevent displacement and enhance stability.

It is also recommended that gabions not be used downstream, as the presence of fluctuating forces in the flow can lead to the dislocation and damage of the wire mesh. In many cases, large stones have been effectively used for protection

and launching, but it has been observed that such stones are often not readily available. In their absence, smaller stones are sometimes used, which tend to wash away, causing the dislocation of the originally designed stones of the required size and weight. This interaction between hydraulic phenomena and structural materials is important for informing the hydraulic design. Since the behavior of these materials in response to dynamic forces is complex and situational, no specific dimensions have been provided in the above discussion, as each case requires a tailored design based on site-specific conditions and material availability.

5. Prototype experiences

Prototype experiences related to the hydraulic performance issues mentioned above can be illustrated through examples such as the Chashma Barrage on the Indus River in Pakistan, which has faced significant degradation since its construction in 1971. The low tail water levels have led to impaired performance of the stilling basin. Historical floods have shown that tail water levels are consistently lower than designed, affecting hydraulic jump formation and causing increased erosion downstream. Furthermore, flood discharges have exceeded the designed capacity, accelerating the degradation of the stilling basin. The insufficient tailwater levels hinder the proper formation of hydraulic jumps, thereby compromising energy dissipation and contributing to the ongoing erosion and structural wear as seen in Photo 2.

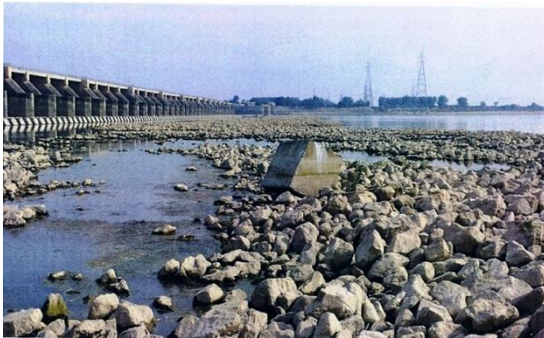


Photo 2: Concrete blocks carried away by river

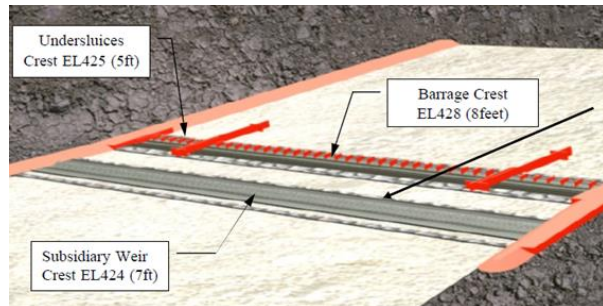


Figure 2: Plan and cross section through under sluices of Tanusa barrage

The study by Chaudary and Sarwar (2014) evaluates the rehabilitation of the Taunsa barrage in Pakistan, focusing on the construction of a sub-weir downstream, which was intended to manage the retrogression by raising the tail water level. However, the findings reveal significant concerns regarding the performance and hydraulic functioning of the barrage and the sub-weir as shown in Figure 2. The 2010 flood exposed vulnerabilities, leading to a breach in the left marginal bund. Despite significant expenditures on rehabilitation, it was noted that the barrage's discharge capacity was compromised, as a sub-weir affected the performance of undersluices. Sedimentation issues between the barrage and sub-weir were overlooked, leading to compromised functionality of the undersluices.

Balegaon barrage is constructed across river Godavari near Nanded in Maharashtra, India. Considerable damage to the river portion, just downstream of the Energy Dissipation Arrangement of the barrage, post construction was observed as seen in Photo 3.



Photo 3 & 4: Damages downstream of energy dissipation arrangement of barrages

Three barrages namely, Medigatta, Annaram, and Sundilla, which are part of the Kaleshwaram Lift Irrigation Scheme in Telangana state of India, faced various issues at downstream. Damages were noticed in 2019, including displacement and settlement of cement concrete (CC) blocks and wearing coat. Remedial measures were delayed, leading to the sinking of piers in Medigatta barrage in 2023. These barrages realized the inadequate tailwater levels leading to high velocities and resultant damages, such as erosion of the wearing coat and displacement of concrete

blocks. Also, frequent operation at small gate openings results in deficient tailwater levels, leading to excessive downstream velocities of the order of 8-10 m/s, that cause structural damages. The lack of actual tailwater rating curve hampers effective discharge estimation and flood management strategies.

6. A case study

Among the numerous case studies of barrages, this paper focuses on one particular example of Annaram Barrage, Kaleshwaram Lift Irrigation Scheme, Telangana. Annaram barrage is having maximum design flood of 65,000 m³/s and standing on the permeable strata made up of stiff clay and sand. The barrage consists of 66 spans of 15 m width having radial gates and 12 m depth of overflow over the crest at Full reservoir level. The stilling basin is provided as energy dissipator for a length of 45 m. There is an impervious barrier made of Secant piling on upstream side as well as downstream side. RCC Raft (M25) had a wearing coat of (M30) 300 mm thick. There was a provision of CC Blocks of size 1.5 x 1.5 x 0.9 m at downstream of end sill for the length of around 50 m. Downstream to that, launching apron in form of loose stone protection of 1800 mm thickness was provided for length of around 60 m. The boulders were laid over the sand bed and packed.



Photo, 5, 6 & 7: Damages to wearing coat and c.c. blocks

During the initial operations of barrage, tailwater levels were almost nil and gates were operated at crack opening which led to high flow velocities and instantaneous damages to the downstream. Due to non-availability of tailwater level, hydraulic jump could not be developed, and energy could not get dissipated. The high energy flow passed downstream and high turbulence was created making damages to the downstream C.C. Blocks and boulders were also dislodged and washed away. The probable causes for the damage of the wearing coat further include high velocity flows, associated turbulence and fluctuating velocities and various joints across the wearing coat. Due to hydraulic forces and consequent turbulence, the

fluctuating velocities were around 2 to 2.5 times the mean values depending on the turbulence intensity at that location. The drag and lift due to these phenomena resulted into the tearing and damages to the wearing coat. The project and design authorities recognized that recent damage to the barrage was primarily caused by inefficient energy dissipation and persistent high downstream velocities, due to low tailwater levels during smaller floods (10,000-15,000 m³/s). This also indicated a time lag in the development of tailwater levels.

Infraplan Hydraulic Laboratory (IHL) was involved in various aspects of the barrage's performance and the hydraulic analysis of damage to the wearing coat and downstream protection works for the project. After site visit and subsequent to numerous discussions with project authorities and after considering several alternatives, IHL recommended the provision of an extended stilling basin downstream of the existing structure. Consequently, the physical hydraulic model studies were referred to IHL for studying energy dissipation arrangement. A geometrically similar 1:45 scale sectional model covering four spans, conforming to Froude's similitude, was constructed at Infraplan Hydraulic Laboratory (IHL) in Pune to assess the efficacy of the existing stilling basin and to study alternative proposals for improving energy dissipation. The studies also included assessments of water surface profiles, flow conditions, the discharging capacity of the barrage, including the partial operation of the radial gates, and evolving the schedule of gate operations.

6.1. Model Studies

A detailed study was conducted at IHL focused on the sectional two-dimensional model of barrage by reproducing three complete spans and two half spans. The goal was to evaluate the hydraulic performance and energy dissipation efficiency of the barrage. Many alternatives to improve energy dissipation efficiency of the barrage were considered such as baffle blocks, raised end sill and extended stilling basin. It was observed that baffles are subject to cavitation damages, in the course of time because of the high velocities at the downstream of barrage. The other alternative

considered was raising the existing end sill from EL 104.5 m to EL 106.0 m. It was observed that, in the absence of tailwater levels, ski-jump like flow conditions were observed in the stilling basin as the downstream tailwater levels upto EL 107 to 108 m were not effective in formation of proper hydraulic jump in the existing stilling basin. Here it may be mentioned that, the length and the depth of the existing stilling basin is inadequate for dissipation of energy as per the design standards and criteria given in BIS in view of the actual tailwater levels realised at the project site.

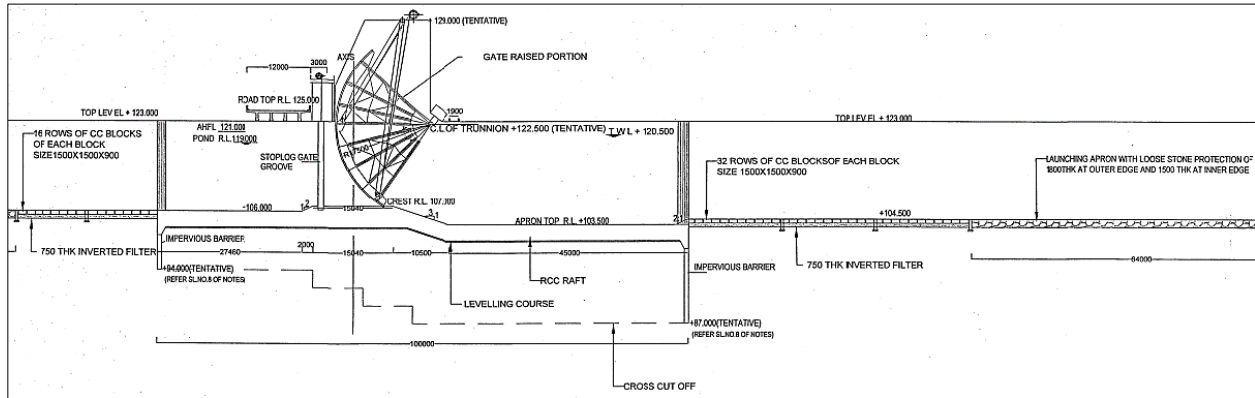


Figure 3: Longitudinal section showing existing barrage components

In view of the above-mentioned hydraulic aspects, it was proposed by IHL to build an extended/secondary stilling basin downstream of the existing end sill with a lowered apron and a higher end sill. Lowering the apron elevation to 100 m was considered suitable, and the end sill height was proposed to raise till 106 m, balancing the flow and minimizing submergence of under-sluices. The extended stilling basin, combined with a vertical wall at the end sill, would protect the structure from undermining and generate higher tailwater levels for smaller discharges. Additionally, submerged divide walls along the centreline of the double piers in the stilling basin were proposed to manage non-uniform flow from the barrage gates, preventing return eddies.

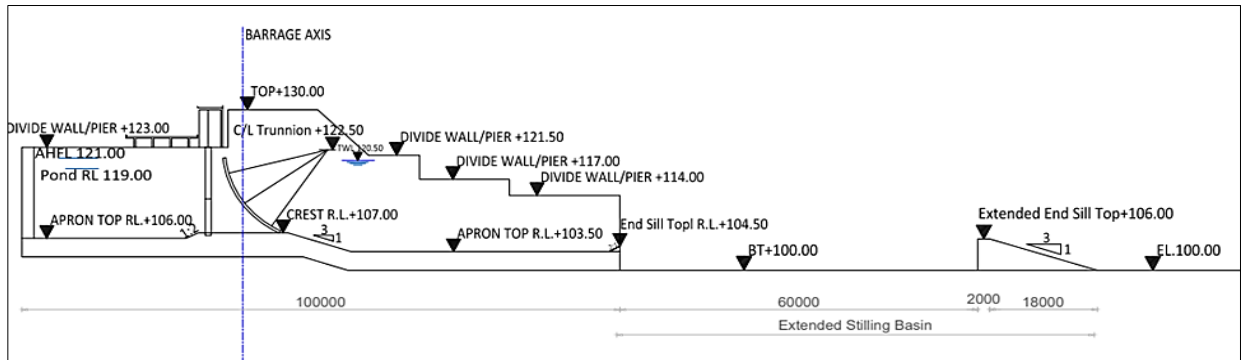


Figure 4: L-section showing proposed extended / secondary stilling basin downstream of existing end sill



Photo 8: Model for existing situation



Photo 9: Model showing extended stilling basin with raised end sill

The hydraulic performance of the stilling basin was tested by measuring velocities, water surface profiles, pressures, and flow conditions. The studies indicated satisfactory performance of both the existing and extended stilling basins across a range of discharges. During initial overflows, velocities were low, but as the head increased, tailwater levels also rose, ensuring effective energy dissipation in the extended stilling basin. For smaller gate openings and low tailwater levels, ski-jump flow occurred due to the existing end sill, but this flow was dissipated effectively without high-velocity impacts reaching the bottom. However, some turbulent flows with velocities of 3 to 5 m/s were observed just downstream of the end sill. When tailwater levels exceeded 110 m, performance improved significantly. For gate openings of 3.5 m and above, it was recommended to open 60 spans equally, leaving three spans on either side closed. Some turbulence and fluctuating flows were observed in the existing stilling basin due to insufficient tailwater levels. Further the studies were carried out to assess the sediment deposition and movement in the stilling basin.



Photo 10 : Flow conditions for lower gate openings



Photo 11:– Flow conditions after addition of extended stilling basin (hydraulic jump formed due to raised end sill of extended basin)

The hydraulic performance of the extended stilling basin was acceptable in terms of protecting the existing structure and diverting the high velocity flows of the order of 8 to 15 m/s downstream of the existing stilling basin by length of 50 m, thereby reducing the velocities in the range of 3 to 5 m/s. It was suggested to provide submerged divide walls in the stilling basin along the centerline of double piers. The studies indicated that the provision of the extended stilling basin would provide sufficient factor of safety to the existing structure as far as hydraulic performance as observed in the model for entire range of discharges and gate openings.

7. Conclusion

This paper highlights the key challenges and design considerations associated with high-head barrages, emphasizing the importance of proper hydraulic design for energy dissipation, so also downstream protection works, and overall structural stability, which are not included in most of the guidelines and design procedures. The study suggests that conventional methods, such as gabions and smaller stones, may not be effective under high-velocity flow conditions, and more robust solutions are required, including larger stone blocks and carefully designed crest geometries.

The paper further emphasizes the use of need for extensive hydraulic model studies to optimize gate operations, stilling basin designs, and downstream protection systems. By incorporating empirical data and prototype performance observations, designers can develop more efficient and resilient structures that mitigate erosion, enhance energy dissipation, and ensure the long-term functionality of high-head barrages. Ultimately, this research highlights the necessity of tailoring barrage designs to site-specific conditions, ensuring that hydraulic performance and safety are prioritized in flood-prone and alluvial river environments.

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